

NYU, Tandon School of Engineering

Bridge to CS

Homework #16

Lab Autograder Note

As this is a lab, the autograder is very general and is **not** meant to indicate full completeness. It provides only a basic evaluation that an attempt was made.

Question 1: Symbol Balance Checker

Implement a symbol balance checker function for the **Pascal** programming language. Pascal allows for the following matching symbol pairs:

- `{ }`
- `( )`
- `[ ]`
- `begin ... end`

All programs will begin with the word `begin` and end with the word `end`.
Your function should:

- Receive an `istream` object which is already open.
- Return `true` if all symbols match correctly, or `false` otherwise.

You do **not** need to worry about comments in the program, but you **must** avoid other parts of the program's code such as assignment statements (e.g., `x = y`) and other expressions.

Examples

- `begin [ ] end` → `true`
- `begin [ ( ] end` → `false`

Please submit a **full program** for this question, similar to your previous assignments. Your submission should include a `main()` function that demonstrates your symbol balance checker working correctly.

Question 2: Queue Using a Vector

Although a queue is best implemented using a list, it can also be implemented using a **vector** if you carefully track the starting position of the queue.

For example:

- If five elements are pushed onto the queue, the start is at position 0 and the end is at position 4.
- If two elements are popped, the start moves to position 2 and the end remains at position 4.
- The popped elements are not physically removed from the vector, avoiding an $O(N)$ pop operation.

Implement a class that uses a **vector** to store the queue. Be mindful of performance, and ensure that if the queue becomes empty, the underlying vector is **reset**.